

Stage 1

Structure-Aware Representation

(2020–2022)

Pre-training + Structure Encoding

Table Image → Structure Parser
→ Row/Column Embeddings
→ Joint Representation → QA Output

*OmniTab · FORTAP · ReasTAP
AIT-QA · FeTaQA*

*Generative
Enhancement*

Stage 2

Generation Enhancement & Modular Reasoning

(2023–2024)

LLM Integration + Decompose-then-Reason

NL Question → LLM Decomposer
→ Sub-queries → Table Executor
→ Answer

*Table-GPT · TableGPT2 · Chain-of-Table
CABINET · TabSQLify · MFORT-QA*

*Verifiable
Execution*

Stage 3

Verifiable Reasoning & Multimodal

(2024–2025)

Execution-Driven + Interpretable

Question → Plan-of-SQLs
→ SQL Executor
→ Verified Answer + Evidence

*Table-R1 · TableGPT-R1 · Weaver
OpenTab · TART · TaPERA*